

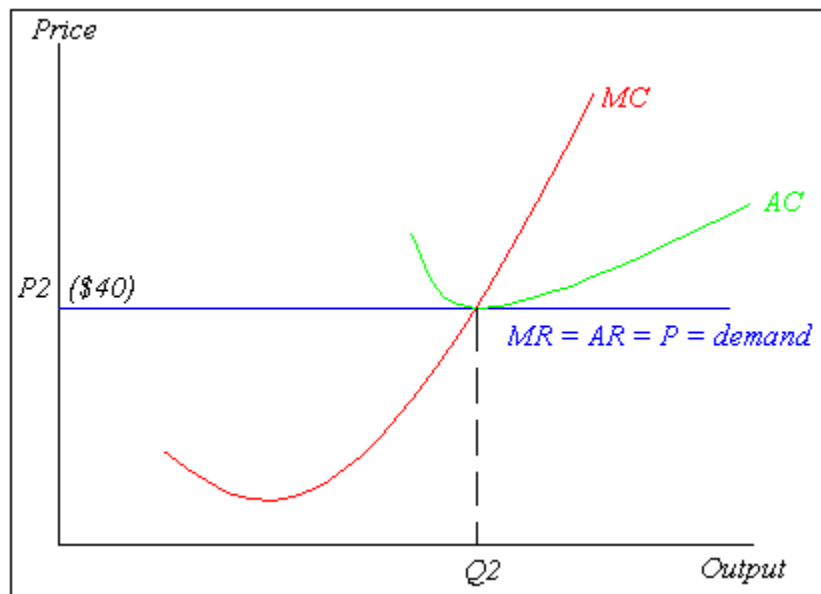
ECO402-MICRO ECONOMICS Lecture Wise Questions and Answers

For Final Term Exam Preparation by Virtualians Social Network

Lecture No 23:

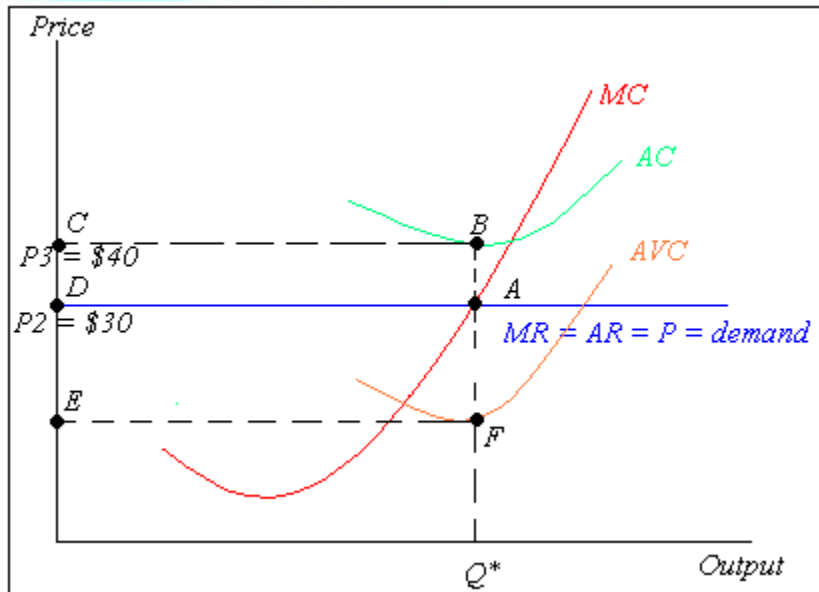
Topic: Perfectly Competitive Markets

Define a competitive firm breaking even?



Firm is producing an output Q_2 , where $MC = MR$ at price (P_2) Its total cost ($AC * Q$) equals its total revenue ($P * Q$) In this case the firm breaks even (or makes normal profit).

What is a competitive firm incurring losses?



At price $P_2 = \$30$ per unit the firm cannot avoid incurring losses, because that price is below the minimum average cost represented by $P_3 = \$40$. At the profit maximizing output q^* the price P_2 is less than an average cost, so that line segment AB measures the average loss from production where $P = MC$, which represent, $AC - P$ (profit/loss per unit). The firm could minimize its losses by producing at q^* , with losses $ABCD$ being incurred, however if the firm were to shut down, it would incur even greater losses equal to the fixed costs of production $CBFE$.

What is Marginal revenue

Marginal revenue is the change in total revenue that firm gets by selling one more unit of output. For example, a firm gets total revenue of Rs. 1000 by selling 10 units of output. This firm gets total revenue equals to Rs. 1200 by selling 11 units of output. The marginal revenue in this case is Rs. 200 which is the additional amount that firm gets by selling additional unit of output.

Lecture No 24:

Topic: The Short-Run Market Supply Curve

What is short run market supply curve?

The short run market supply curve

Amount of output that the industry will produce for every possible price in short run shows short run market supply curve. The industry's output is the sum of the quantities supplied by all of its individual firms. By adding the supply curves of each firm we can get market supply curve. Suppose there are three firms and each firm have different production costs. Supply curve of each firm will be the portion of marginal cost which lies above the average variable cost. Among these three firms, a firm which has low average variable cost and market price is above it, will supply more units as compared to other firms that have high average variable cost.

Define the concept of perfectly elastic and inelastic for short run supply?

If supply of a good is relatively more elastic, it means its production responds/changes quickly due to change in its price in the short run. For example supply of burgers (fast food) is relatively more elastic in the short run. Some goods have relatively more inelastic supply curve in the short run. For example supply of housing is more inelastic. Its supply does not response quickly even though a large change in price has been occurred.

what is the difference between the accounting profit and economics profit?

Accounting profit = Total revenue - Explicit Cost

Economic Profit = Total revenue - (Explicit cost + Implicit cost).

Lecture No 25:

Topic: Equilibrium In Perfectly Competitive Markets

Q. How can we measure the degree of economics of scope, Kindly explain with the help of Formula?

Economies of scope mean output produced jointly by a single firm is greater than output that could be produced by two different firms, when each produces a single product. For example, KFC can produce both Chicken burger and French fries at a lower average cost than what it would cost two separate firms to produce the same goods. This is because KFC's Chicken burgers and French fries share the use of food storage and preparation facilities.

Economies of scope can also be found by studying a firm's costs. If a combination of inputs used by one firm produces more output than two different firms would produce, then the costs for the firm producing jointly will be less than the costs of the firms producing separately. We can measure the degree of economies by following formula:

$$SC = \frac{C(Q1) + C(Q2) - C(Q1, Q2)}{C(Q1, Q2)}$$

SC is degree of economies of scope

C (Q1) is the cost of producing Q1

C (Q2) is the cost of producing Q2

C (Q1Q2) is the joint cost of producing both products

If $SC > 0$ Economies of scope exists

If $SC < 0$ Diseconomies of scope exists.

what is the opportunity cost of land?

Opportunity Cost of the land means, the revenue which can be obtain by the land owner by renting it to some one in spite of using it own.

Is zero profit is necessary or is must in long- run equilibrium.?

In long run the firms get zero economic profit because when firms earn supernormal profit then new firms enter in the market. In this way the supply of the product increases and price come down and the firms earn normal profit. In case of the loss the situation will be reversed.

Lecture No 26:

Topic: characteristics of a market structure

Define what are the characteristics of a market structure?

How to solve following query?

Probably	Return \$
0.2	100
0.4	50
	-25

0.4

Define the Characteristics of market structure?

Characteristics of a market structure

Following are the characteristics of market structure

- Number of firms in the market/industry
- Extent of barriers to entry
- Nature of product
- Degree of control over price.

why AVC ,ATC and MC are U shaped?

These curves have U shape because they indicate that when production increases they start to fall. When these curves reach to minimum point, and if firm produce more then these curves will start to rise.

Lecture No 27:

Explain the following;

a) Marginal Cost and Marginal Revenues?

b) Total Cost and Total Revenues?

Solution :

Marginal cost

Increase in cost resulting from the production of one extra unit of output is called marginal cost.

$$MC = \Delta TC / \Delta Q$$

For example, Total cost is Rs.1000 when first unit is produced and total cost becomes Rs.1500 when second produced so, marginal cost is Rs.500 which is the change in total cost due to change in output.

Marginal revenue

Change in revenue resulting from a one unit increase in output is called marginal revenue.

$$MR = \Delta TR / \Delta Q$$

For example, Total revenue is Rs.500 when first unit is sold and total revenue becomes Rs.1500 when second sold so, marginal revenue is Rs.1000 which is the change in total revenue due to change in output sold.

Total revenue

Total amount received by a firm after selling the whole of the output produced during a given period of time.

$$TR = P * Q$$

Total cost

It is the sum of fixed and variable costs at each level of out.
 $TC = TFC + TVC$

Why elasticity of demand for downward slopping curve?

Downward demand curve may be flatter or steeper. If the demand curve is flatter then elasticity of demand will be greater than unity or we can say curve is more elastic. If the curve is steeper then the elasticity of demand will be less than unit or one in other words we can say demand curve is less elastic.

Lecture No 28:

Topic: The Analysis Of Competitive Markets

What is Tarrif?

Tarrif

Tarrif is import tax which is levied by government to discourage the imports. Imports could be reduced to zero by imposing a sufficiently large tariff. The tariff would have to be equal to or greater than the difference between P_o (domestic price) and P_w (World price). With a tariff of this size, there will be no imports and, therefore, no government revenue from tariff collections, so the effect on consumers and producers would be the same as with a quota. For this purpose government can also use the import quota system.

what does the term quotas mean?

Quota is a limit set on the quantity of a product that may be imported or exported within a given period.

what is Ps?

P_s is price support which is set by government to support the producers. It is usually set above the market clearing price and if government wants to collect amount of tax from producer then it is below the market clearing price.

what is pass through fraction 1?

Pass through fraction formula tells us what fraction of tax is borne by consumer in the form of higher prices.

Pass through fraction = $E_s / (E_s - E_d)$

For example, when demand is totally inelastic, so that E_d is zero, the pass through fraction is 1, and all the tax is borne by consumers. When demand is totally elastic, the pass-through fraction is zero, and producers bear all the tax. (The fraction of tax producers bear is given by $-E_d / (E_s - E_d)$).

Define Cardinal utility and ordinal utility?

Cardinal utility and ordinal utility approaches are used to study about the utility that a consumer gains by the consumption of any good.

Lecture No 29:

What is the calculation of subsidy?

Calculation of Subsidy

how you calculate the subsidy .80

It is assumed that Government has announced subsidy of 80 percent on quantity of output produced. This value is assumed value.

What is Deadweight Loss?

Deadweight loss is the sum of consumer loss and producer loss. According to the example area 'B' is consumer loss which can be calculated as follows
consumer loss = $\frac{1}{2} (1.22 - 1)(89 - 100)$
= $\frac{1}{2} (.22)(-11)$
= 1.21 billion per year (Negative sign indicate loss).

Lecture No 30:

Define barriers , natural and artificial monopoly?

Barriers to entry are the obstacles that a firm faces when it takes a decision to enter into the market. For example, cost of capital, advertisement cost, patents, trade marks etc.

In natural monopoly, cost required to start a business is very high and it's better for society that only single firm operates instead of multiple firms in such an industry. For example, WAPDA is having a natural monopoly.

Artificial monopoly is formed when government impose artificial barriers on different firms to enter into the market like licensing. Once barriers have been removed, new firms enter into the market to compete with monopolist.

Please define Marginal & Average product?

Average product: Average product is the per unit production of a firm.

Average product = Total product/ Variable inputs.

Marginal Product:

The marginal product is the output produced by one more unit of a given input.

Diminishing returns, also called law of diminishing returns, is an economic law stating that if one input in the production of a commodity is increased while all other inputs are held fixed, a point will eventually be reached at which additions of the input yield progressively smaller, or diminishing, increases in output.

Lecture No 31:

What is the cost function?

The cost function is defined as a function of input prices and output quantity.

$C=f(Q)$ where C is cost and Q is quantity produced. As quantity produced increases total cost also increases.

Define Total revenue?

- Total revenue is found by multiplying price and quantity sold.

$TR = P \times Q$ e.g. price Rs.10 per unit and quantity sold is 15 units then we can find total revenue as
 $TR = P \times Q = 10 \times 15 = \text{Rs.}150$

$$P = 6 - Q$$

- $P = 6 - Q$ it's a demand curve which is faced by firm. We can find Total revenue as
 $TR = P \times Q$

$$TR = (6 - Q) Q$$

$$TR = 6Q - Q^2$$

What is Two-part tariff?

Two-part tariff is defined as buyer pays to the seller a lump-sum fee and a constant charge for each unit of the product or service purchased. For example, a credit card carries a two-part tariff if it has an annual fee and a minimum fee with each purchase.

what is the difference between MC and MR?

Marginal revenue is the additional revenue that firm gets by selling one more unit of output. For example, a firm is selling 100 units at price Rs.50 per unit and gets total revenue of Rs. 5000. When this firm sells total 101 units at same per unit price, it gets total revenue of Rs.5050. The marginal revenue in this example is Rs.50.

Marginal cost is the additional cost that firm bears in producing one additional unit of output. For example, a firm is producing 100 units at per unit cost of Rs.50. Total cost of producing 100 units is Rs. 5000. When this firm decides to produce total 101 units at same per unit cost, it bears total cost of Rs.5050. The marginal cost in this example is Rs.50.

Lecture No 32:

Define monopolist's average revenue?

The monopolist's average revenue means the price received by monopolist per unit sold it is also market demand curve. Marginal revenue is the change in revenue resulting from a one unit increase in output. Slope of marginal revenue is twice of average revenue. Marginal revenue is below the average revenue because it's a change in revenue with the sale of one more unit while average revenue is average. Keeping these things you can draw marginal revenue and average revenue.

What is Economic rent and opportunity cost?

Economic rent means amount that firms are willing to pay for an inputs (such as land, capital or labor) less the minimum amount necessary to obtain it.

Opportunity cost associated with opportunities that are forgone when a firm's resources are not put to their highest value use.

Let's suppose the factor of production is labour. In this example, the labourer receives Rs.200/per day for his job, and the minimum salary they'd be willing to work for (opportunity cost) is Rs.160/per day. This Rs.40/day difference is the laborer's economic rent.

Difference between Monopoly and Depreciation?

Monopoly: A situation in which a single company or group owns all or nearly all of the market for a given type of product or service.

Depreciation: Depreciation means decline in the value of assets.

Lecture No 33:

What is Social Cost?

SOCIAL COST: SOCIAL COST is the cost to society as a whole from an event, action, or policy change.

What is marginal and average expenditure and marginal value and marginal revenue?

- Marginal expenditure means additional cost of buying one more unit of good. For example, a person buys 10 units at the price Rs.50 per unit. His total expenditures are Rs.500 if he buys one more unit his total expenditure are Rs.550 and his marginal expenditure is Rs.50 $(550-500/11-10)$.
- Average expenditure means price paid per unit of a good.e.g. A person buys 10 units and his total expenditure is Rs.600 then his average expenditure is Rs.60 $(600/10)$.
- Additional benefit derived from purchasing one more unit of good is called marginal value.
- Change in revenue resulting from a one unit increase in output is called marginal revenue.

$$MR = \Delta TR / \Delta Q$$

For example, Total revenue is Rs.500 when first unit is sold and total revenue becomes Rs.1500 when second sold so, marginal revenue is Rs.1000 which is the change in total revenue due to change in output sold.

What is Rent seeking?

Spending money in socially unproductive efforts to acquire, maintains, or exercise monopoly power is called rent seeking.

What is Monopsony?

A monopsony is a market in which there is a single buyer. Monopsony power is the ability of the buyer to affect the price of the good and pay less than the price that would exist in a competitive market.

what is the value of MC from where it is come?

The calculation can be made in this manner

$$\frac{MC}{0.9} = \frac{1}{0.9} \cdot MC = 1.1 MC$$

We haven't put any value for MC here.

Lecture No 34:

What is Economies of scale?

The reduction in long run unit costs which arise from an increase in production/purchasing is due to the fixed costs being spread out over more products and is called economies of scale.

What is tariffs, bundling and variable profit?

Two-part tariff is a pricing strategy in which the price of a good or service is composed of two parts - a lump-sum fee plus per-unit charge. In this strategy, buyer pays to the seller a fixed fee and a constant charge for each unit purchased. It increases profit of firms. A credit card is an example of two-part tariff. Banks charge annual fee plus per transaction fee of using credit cards.

Bundling is packaging of two or more commodities to gain the pricing advantage. This is the way to increase the profit, by combining only loosely related products as a bundle for sale. This strategy is very common in software business. Like giving away free floppy disk with the purchase of computer.

Profit is the difference between total revenue that firm gets by selling quantity of output and total cost of producing that output.

What is price discrimination?

Price discrimination means to sell a product in different markets at various prices. If we sell same product in Lahore and Karachi but at different prices then it will be called price discrimination.

Lecture No 35:

What is incremental profit explain with an example?

Incremental Profit:

Incremental profit is the additional profit that firms get by selling additional unit of output. For example, firm A is getting total profit of Rs. 10,000 by selling 100 units. When it sells 101 units, it gets total profit of Rs. 10,050. The incremental profit in this example is Rs.50 which firm gets by selling one more unit.

This is an example which explains the concept of differences in elasticity for users and non-users. Price elasticity of demand of any particular product for non-users is always greater than price elasticity of demand for users. This is because users of product will not leave consuming that product when its price increases. So, in case of users, change in quantity demanded is less than change in price (low elasticity). In the given example of cake mix, it is assumed that price elasticity of demand for users and non-users is -4 and -2 respectively. By putting these values in given formula, we have found that price charged to non-users is 1.5 times higher than price charged to users as their elasticity of demand is relatively high. You can take any values of elasticities but Price elasticity of demand of any particular product for non-users should always be greater than price elasticity of demand for users.

What is the Types of Price Discrimination:

First Degree Price Discrimination:

In first degree price discrimination, different prices are charged to different customers according to their willingness to pay. Sellers charge separate price to each customer based on their willingness to pay. Examples; Lawyers charge fee from customers according to their ability to pay.

Second Degree Price Discrimination:

In second-degree price discrimination, price is charged according to quantity consumed. For example, discount will be given to customers who purchase quantities in bulk.

Third Degree Price Discrimination:

In third degree price discrimination, sellers charge different prices of a product from different consumer groups based on elasticities of demand. They will charge higher price to consumers having more inelastic demand curve and lower price to consumers having more elastic demand curve. For example, business air travelers versus vacation air travelers.

Lecture No 36:

What is Trade off?

Trade off

A technique of reducing or forgoing one or more desirable outcomes in exchange for increasing or obtaining other desirable outcomes in order to maximize the total profit or efficiency under given circumstances is called trade-off. For example if the management of amusement park wants to maximize total profit they charge high entrance fee and less fee for each ride so there is trade-off between entrance fee and ride fee. According to the statement total profit is the sum of the profit from the entry fee and profit from sales depends on 'T' which is entry fee, if entry fee will be less charged more visitors will come to amusement park and they will use equipment in the park and if entry fee will high then there will be less number of entrants in the park that is why profits depend on 'T'.

What is Rebate?

A partial refund of the original payment for some services or purchasing goods is called rebate. For example, Kodak runs a program in which a consumer can fill a form and can get rebate Rs.25 on the purchase of film roll.

Define Rule of Thumb?

- Similar demand: Choose P close to MC and high T
- Dissimilar demand: Choose high P and low T.

Lecture No 37:

Define Hetrogenous Customers?

Hetrogenous customers mean different customers having different income, taste and preferences.

What is Leakage?

Leakage is the amount of money which flows out of the circular flow of income. It is part of income (received by consumers in return of services which they provide) that consumers don't spend on purchasing goods and services. This income is received by the consumers but not returned to the firm. For example, savings, imports, taxes etc.

Define Real GDP?

Real GDP is the inflation adjusted value of total output produced within boundary of the country over a given time period. It is calculated by adjusting nominal GDP with inflation.

Lecture No 38:

How quantity demanded depends on price and advertising?

When price increases then quantity demand decreases so quantity demand has negative relation with price.

When there will more advertisement for any good then people will like to use that product and hence it will increase quantity demand.

What is stock variables?

Stock variables are wealth, government debt, capital, number of people losing their jobs etc.

When The long run aggregate supply curve will shift to the right?

Long run aggregate supply curve changes due to following factors;

- change in factors of production,
- change in technology,
- change in price of something needed for production like oil etc.

What is the types of Bundling?

- Mixed bundling: Selling both as a bundle and separately
- Pure bundling: Selling only a package

Lecture No 39:

What is Explicit collusion and implicit collusion?

An explicit collusion is a formal, usually secret, collusion agreement among competing firms in an industry designed to control the market, raise the market price, and otherwise act like a monopoly. Explicit collusion is also known as overt collusion. The distinguishing feature of explicit collusion is a formal agreement whereas no formal explicit agreement is involved in implicit collusion.

What is Cournot equilibrium?

Cournot equilibrium occurs where the reaction curves of the two firms intersect. It is used for the production analysis.

Define Reaction curves?

Reaction curves are used to see the production of one firm by keeping eye on the other firm. Reaction curve shows the total output produced by the firms.

What is Complement goods

Complement goods can be consumed collectively. Cross price elasticity of demand is negative in case of complements which show that quantity demanded of good decreases in response to increase in price of its complement good.

Lecture No 40:

What is Oligopoly?

Oligopoly is a market structure dominated by a small number of firms, selling, either identical or differentiated products or significant barriers to entry into the industry.

What is Nash Equilibrium?

Each firm is doing the best it can given what its competitor are doing.

What is stackelberg Model and Bertrand Model?

In stackelberg model there must be a dominant firm who took the initiative to setting the price keeping eye on the reaction of the other firm, while in the Bertrand model we are taking about positive Marginal cost and homogeneous products.

Which factor increase when output increase?

In the short run, some factors of production are fixed (like plant, machinery etc.). Firm increase output by increasing variable factors of production (like labors, raw material etc). In the long run, all factors of production are variable. Firm increase output by adjusting both fixed and variables factors of production.

Lecture No 41: Define Cartel?

Cartel is an organization of similar, independent companies which join together to take output and price decisions that maximizes their joint profits. The entire industry will follow the guidelines set by the Cartel. All cartel members will not compete with each other and follow the mutual decision taken by cartel. For example, Organization of Petroleum Exporting Countries (OPEC). It will take all price and output/supply decisions and all oil producing countries (which are members of OPEC) will follow these decisions.

What is Oligopoly?

Oligopoly is a market structure dominated by a small number of large firms, selling either identical or differentiated products, or significant barriers to entry into the industry. This is one of four basic market structures. In oligopoly, firms make production decisions keeping in view the actions of other firms. For example, Cement, cars, electrical appliance.

What is Variance?

Variance measures deviation/dispersion of outcomes from average expected outcome. It measures the risk that an investor might take on when purchasing a specific security. Its value is either zero or positive. Higher variance shows higher risk that an investor takes on.

Why is the long-run average cost (LRAC) curve U-shaped?

Long run average cost curve is a u shaped curve showing that as firm increases output, his long run average cost declines initially due to economies of scale (productive efficiency, better utilization of resources etc). If he continues to increase his output, he neither experiences economies nor diseconomies. Finally, with more production, firm experiences diseconomies of scale.

Lecture No 42:

Define the factors of productions?

There four factors of production:

1. Land
2. Labor
3. Capital
4. Entrepreneurship

What is Real exchange rate?

The relative price of domestic goods in terms of foreign goods is called real exchange rate. Real exchange rate $= E = e * P / P^*$, where E is the real exchange rate, e is the nominal exchange rate, P and P* are domestic and foreign prices respectively

Lecture No 43:

Why is the long-run average cost (LRAC) curve U-shaped?

Long run average cost curve is a u shaped curve showing that as firm increases output, his long run average cost declines initially due to economies of scale (productive efficiency, better utilization of resources etc). If he continues to increase his output, he neither experiences economies nor diseconomies. Finally, with more production, firm experiences diseconomies of scale

When demand is elastic and inelastic?

Demand is said to be inelastic when percentage change in quantity demanded of a commodity is less than percentage change in price. Consumers will not change (increase or decrease) consumption of that commodity due to change in its price. People will not cut demand due to increase in price.

Demand is said to be elastic when percentage change in quantity demanded of a commodity is more than percentage change in price. A small change in price will bring a large change in quantity demanded of that commodity.

Differentiate between shortage and surplus?

Shortage is condition in the market where quantity demanded (Q_d) is greater than quantity supplied (Q_s). Surplus is a condition where quantity supplied is greater than quantity demanded. Suppose quantity demanded of good is 100 units while quantity supplied of that particular good in the market is 80 units. It shows a shortage of 20 units in the market.

Can perfectly competitive firms earn economic profit?

An economic profit arises when revenue (amount that firm gets by selling output) is greater than all costs (that firm incurs in producing output). It is not possible that perfectly competitive firm earns economic profits in the long run. If firms earn economic profits, then new firms enter in to the market which increases competition among firms. Entry of new firms is continued till all firms get normal profits.

Lecture No 44:

Define economic rent?

For a factor market, economic rent is the difference between the payments made to a factor of production and the minimum amount that must be spent to obtain the use of that factor.

Lecture No 45:

What factors determine output level under monopoly

Firm operating under monopoly will produce output at a level where its profits are maximized. Profit maximizing level of output will be determined at a point where firm's marginal cost equals marginal revenue.

The price elasticity demand for demand curve zero that also 0 or not?

On negatively sloped demand curve, there are different values of elasticity of demand. It is less than one on all points lying below mid point and greater than one on all points lying above mid point. Its value is 1 at mid point. In case of perfectly inelastic demand curve, elasticity of

demand is zero which shows that quantity demanded of a commodity does not change even a large change in price has been occurred

What is Substitute goods?

Substitute goods are those that can be used alternatively. Cross price elasticity of demand is positive in case of substitutes which shows that quantity demanded of good increases in response to increase in price of its substitute good.